

Learning outcomes	Additional guidance
<i>Learners should be able to demonstrate and apply their knowledge and understanding of:</i>	
(a) Einstein's mass–energy equation; $\Delta E = \Delta mc^2$ (b) energy released (or absorbed) in simple nuclear reactions (c) creation and annihilation of particle–antiparticle pairs (d) mass defect; binding energy; binding energy per nucleon (e) binding energy per nucleon against nucleon number curve; energy changes in reactions (f) binding energy of nuclei using $\Delta E = \Delta mc^2$ and masses of nuclei (g) induced nuclear fission; chain reaction (h) basic structure of a fission reactor; components – fuel rods, control rods and moderator (i) environmental impact of nuclear waste (j) nuclear fusion; fusion reactions and temperature (k) balancing nuclear transformation equations.	HSW9, HSW10, HSW12 Decision making process when building new nuclear power stations. THIS WILL ALSO REQUIRE KNOWLEDGE OF 5.1.4